

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 3 – MCQ Test

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO3 – Precision	Assessment:	Assessment Tool 3 – MCQ Test
Weightage:	30% of CIA		
CO5 Statement:	Apply N-gram models, smoothing techniques, part-of-speech tagging systems and to evaluate effective syntactic analysis. (BL-3)		
Student Name:		Reg. No.:	
Section / Batch:		Date:	

Code of Conduct

I B Pranith Sai Nihal(192472294) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: PRANITH SAI _____

Instructions

- Read each question carefully before selecting your answer.
- This assessment consists of 50 multiple-choice questions.
- Do not use external resources, AI tools, textbooks, or assistance from others during the assessment unless explicitly permitted by the instructor.
- Upload Score Card in GCR and submit score card printout.
- Duration: 30 minutes.

Section / Topic		Focus	Q Nos
Context-Free Grammars (CFG)	CFG components, productions, terminals, non-terminals, recursion, sentence generation		1 – 10
Sentence Construction and Agreement	Parse trees, sentence structure, grammatical agreement, syntactic validation		11 – 20
Subcategorization	Verb frames, complement requirements, argument structure analysis		21 – 25
Top-Down Parsing	Parsing strategy, derivation process, recursion issues, grammar expansion		26 – 30
Earley Parsing	Predictor, Scanner, Completer operations, ambiguity handling		31 – 35
Feature Structures	Feature representation, unification, constraint checking, agreement enforcement		36 – 40
Probabilistic Context-Free Grammars (PCFG)	Rule probabilities, ambiguity resolution, probabilistic parsing		41 – 48
Integrated Syntax Analysis	CFG + Feature Structures + PCFG integration in NLP applications		49 – 50

Annexure A – MCQ Test

<https://forms.gle/WSv28zfaU5B2aWcy6>

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Number of Correct Answers	100%	45 to 50 correct answers	38 to 44 correct answers	25 to 37 correct answers	0 to 24 correct answers
Accuracy	100%	Excellent (90–100%)	Good (75–89%)	Average (50–74%)	Poor (Below 50%)

No. of Questions	Number of Correct Answers	Accuracy	Total %
50			

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

subject - NLP

MCA-test

B. Prarith Sai Nihal

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- 1) Recursive product rules allow hierarchical sentences - B X
- 2) Subject-verb Agreement - A ✓
- 3) finite state of Argument - C X (17)
- 4) Morphology - A X
- 5) Left recursion X
- 6) uses predictor, scanner, and Completer operations to process partial input efficiency - B ✓
- 7) Regular expression matching - A X
- 8) chooses the parse tree with the maximum rule probability - A ✓
- 9) DFA - Regular expression - A ✓
- 10) Regular expression → B X
- 11) removed Remove all → B ✓
- 12) The list of authorized employees are updated - C X
- 13) The verb violates its frame - C
- 14) Left recursion causing infinite recursion - B
- 15) Predictor - B X
- 16) Number feature conflict between subject and verb - A X
- 17) Tree 2 because it has higher probability - A X

- 18) High recursive grammar with unrestricted productions - A X
- 19) Earley parsing handles ambiguity effectively - B X
- 20) feature structures $\rightarrow$ CFG $\rightarrow$ Earley parser - Tokenization - B X
- 21) Add the production N - "living room lights" - B X
- 22) Main subject noun phrase - A X
- 23) verb subcategorization frame - D X
- 24) Lexical Ambiguity - B ✓
- 25) Scanner - B X
- 26) maximum likelihood estimation - B X
- 27) Tree A because it has high probability - A ✓
- 28) Context-free - Grammar only - A X
- 29) Recursive Top-Down parser - B ✓
- 30) Highest PCFG rule probability - D X
- 31) $S \rightarrow NPVP \rightarrow Det NP \rightarrow the robot VP - VNP - derives \rightarrow A ✓$
- 32) The Engineer develop the software - B X
- 33) she gave the customer a refund - C X
- 34) Expand VP - B X
- 35) Completer - C X
- 36) fails - A ✓
- 37) the shortest parse tree - A ✓

So far with 50
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38) Recursive descent parsing - B X

39) (A) Parse A because it satisfies the constraints and has higher probability.

40) Accept the parse because CFG validation succeeded - B

41) $VP \rightarrow VNP \rightarrow A$

42) $N \rightarrow C X$

43) Scanner $\rightarrow B \checkmark$

44) Sentences accepted because person matches - A $\checkmark$

45) the max number of production rules - A X

46) CFG support recursive noun phrase - A $\checkmark$

47) structural ambiguity (C) $\checkmark$

48) Left recursive - B $\checkmark$

49) The verb requires additional complements - A X

50) $VVP PP - C X$

